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» Tutoriales / Algoritmo A* para encontrar el camino más corto en IA

Algoritmo A* para encontrar el camino más corto en IA

Por Hernán el 04 de Octubre de 2010 con 61,122 visitas
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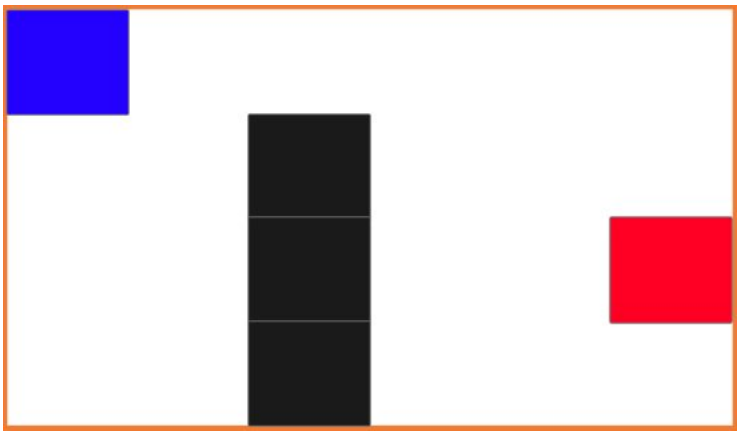
El algoritmo Pathfindig A* (**A estrella** o **A asterisco**), es el más conocido de todos para encontrar el camino más corto entre un punto dado (A) y otro (B). Es muy usado en juegos, el primero en usarlo fue el famoso Pac-Man (Para los fantasmitas).

El método que explicaré esta "*algo*" retocado por mí, más que nada porque generalmente lo he usado en ActionScript y he querido mejor el rendimier Aunque esta MUY poco tocado, básicamente simplifique los valores de la fórmula y edité el uso de listas abiertas/cerradas para no guardar basura en RAM.

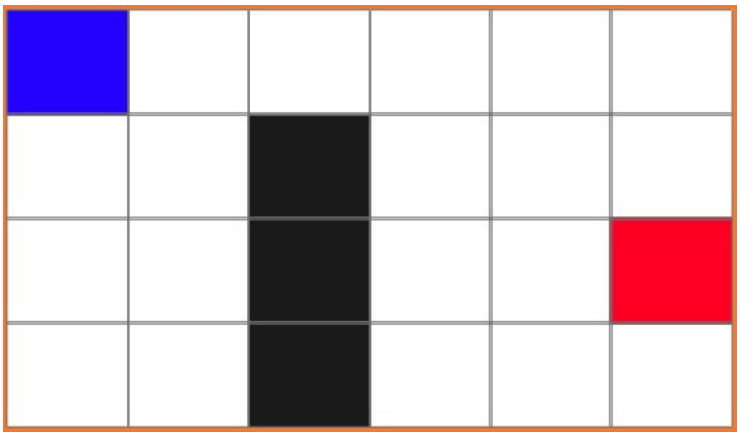
No nos enfocaremos en código sino en la comprensión del Algoritmo. Hay algunos ejemplos de todas formas aquí en Clab.

Escenario base para el algoritmo A estrella o A asterisco

Imaginen tenemos un mapa o escenario, dónde encontraremos dos puntos A (Azul) y B (Rojo). Así como obstáculos (Negro).



Lo primero debemos hacer es representar este mapa a nivel lógico, por ello lo dividiremos en una cuadricula:



Cada cuadrado en la cuadricula es un punto de movimiento (El movimiento mínimo que podremos obtener). El sistema realmente puede usar cualquie geométrica poligonal. Los cuadrados son más fáciles.

Convertir nuestro escenario o mapa a datos

Ahora el problema es como representamos a nuestro código una cuadricula. Pues en realidad fácil, con una cuadricula o Array, donde señalaremos 0 espacio vació, 1 espacio ocupado.

Para mi ejemplo:

Código :

```
[0,0,0,0,0,0]
[0,0,1,0,0,0]
[0,0,1,0,0,0]
[0,0,1,0,0,0]
```

Notarán no puse los puntos de espacio de A ni B, dado que podrían ser dinámicos. Esos los manejaremos con posición:

Código :

```
A = [0,0]
B = [5,2]
```

Algoritmo PathFinding A* explicado para humanos

La formula del Pathfinding A* es realmente simple:

Código :

```
Movimiento = Esfuerzo + Heurística
```

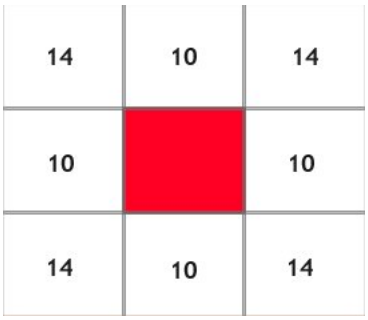
Primero que nada, debemos entender que solo podemos movernos hacia 8 direcciones: 2 Horizontales, 2 Verticales (Valores Ortogonales) y 4 Diagor

Cada movimiento le asignaremos un valor definido (El cálculo real usa raíces, pero no consume bastante más):

Código :

```
Valor Ortogonal = 10
Valor Diagonal = 14
```

Así que el esfuerzo será determinado según el movimiento que haremos, Ortogonal u Horizontal:



La variable *Heurística* en cambio, es una variable "a ojo". Como no sabemos el valor real, calcularemos la distancia entre A <-> B[Análisis] desde sus Ortogonales sin obstáculos. Lo cual es fácil, ya que sabemos la posición en la matriz de ambos objetos.

Código :

```
Heuristica desde Punto B (Inicial) = 7 * 10
```

O lo que es más correcto:

Código :

```
Heuristica = Distancia desde el punto de Analisis * 10
```

Hemos sumado 10 ya que tenemos una incertidumbre incierta, esto debería cubrirnos.

Este método de calcular la Heuristica se llama Manhattan.

Mejorar el rendimiento del Algoritmo A* estrella/asterisco

Siempre que uso el Pathfinding A* debo admitir no lo hago exactamente así, ya que nunca me ha gustado calcular 8 direcciones, cuando se para qué dirección quiero ir (Derecha o Izquierda).

Generalmente lo que hago es "eliminar" las posiciones de cálculo que no corresponden al lugar donde quiero ir. Esto lo hago calculando si mi posición B hacia A es negativa o positiva:

Código :

```
Direccion = Posicion B Horizontal - Posicion A Horizontal
```

Esto elimina una buena parte del cálculo (Ahora veremos porque esto nos preocupa).

14	10	14
10		10
14	10	14

Calcular el camino más corto

Antes que nada, debemos saber que calcularemos a la inversa, desde el punto B hacia el A (Por practicidad). Es lo mismo el path de ida que de venic que no afectará nada.

Para calcular debemos crear una progresión usando la fórmula que ya sabemos. Nos paramos en la posición B, y calculamos cual es el valor de B qu nos conviene tomar según la fórmula que ya conocemos.

Código :

```
Valor = G + H
```

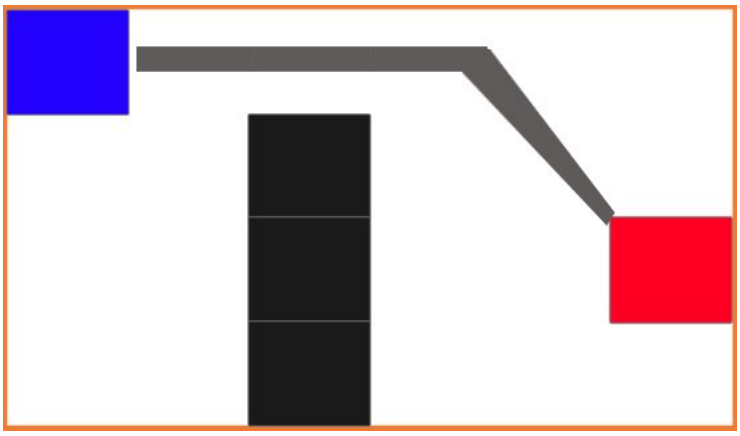
G = 14 H = 5 Valor = 19	G = 14 H = 6 Valor = 20	Anulado
G = 14 H = 6 Valor = 20		Anulado
G = 14 H = 7 Valor = 21	G = 14 H = 8 Valor = 22	Anulado

Nota: Quité la multiplicación, porque funciona en valores cortos bien y no me pareció necesario para ejemplificar la idea. En cuadriculas grandes emplear la multiplicación.

Deberán ejecutar la fórmula pero evaluando que el cálculo sea posible, si el cuadrado es 0 (libre) o 1 (ocupado). En caso de 1 no valdrá el camino. In pueden ejercer una regla, donde la diagonal no sea posible si su ortogonal esta ocupada (Es a criterio de cada uno).

Camino Final calculado por A*

Obviamente calcular el camino final implicará ir eligiendo el valor más corto y desde allí replicar la fórmula hasta llegar al punto A. Luego obviamente la lista.



Resultado:

Código :

```
Movimiento 1 = [1][0]
Movimiento 2 = [2][0]
Movimiento 3 = [3][1]
Movimiento 4 = [4][2]
```

Aplicaciones del Pathfinding A*

Este algoritmo de Inteligencia Artificial es muy útil para calcular el camino óptimo entre el punto A y B. Aunque también nos sirve para estimar compar Si sabemos cual es el valor de todo el movimiento del camino y sabemos el valor del camino que hizo el usuario, podemos estimar cual fue su eficien

Si lo que queremos usar es mapas, este Algoritmo debe ser aplicado mediante Waypoints.

Espero les sirva 😊

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Por Freddie

Por Hernán el 04 de Octubre de 2010

Mierda, re-leyendo me acabo de dar cuenta que me olvidé de aclarar que la dirección se debe "habilitar" en caso de que las otras 5 respuestas no tengan solución, y que de ser así, debemos registrar las soluciones del paso anterior para no "ir para atrás" cuando estamos usando dirección no acorde a la más lógica (Laberintos aplica esto).

Saludos, Hernán . -

Por Zah el 04 de Octubre de 2010

Muy interesante .
Cabe señalar que la idea se puede extender de muchas maneras, como por ejemplo asignándole un valor diferente a la distancia según el tipo de camino que se tratase (para que por ejemplo no sea lo mismo desplazarse por carretera que por bosque. Quedaría:

Código :

```
distancia = multiplicador_de_dirección * velocidad_de_la_baldosa
```

donde multiplicador_de_dirección es 1 para las direcciones paralelas y raiz de 2 (1.4) para las diagonales.
y velocidad_de_la_baldosa puede depender del tipo de terreno, del objeto que se esté moviendo, de si hay transición entre terrenos distintos, o incluso de si la IA estima que es más o menos probable encontrarse con enemigos.

Saludos.

Por Hernán el 04 de Octubre de 2010

Oh si Zah, tienes razón... Incluso se podría conectar con el tip anterior de GPS y escribir un rutina de los caminos que según ciertas pautas sean más seguros o más rápidos de recorrer para el tipo de vehiculo que tenga en el momento, enconjunto a la formula de distancia que dices.

Por Dano el 04 de Octubre de 2010

Se ve bien, muy didáctico.

Si un día tengo un tiempo muerto por ahí, veré si hay maneras de optimizarlo.

Por Hernán el 04 de Octubre de 2010

Dano :

> Se ve bien, muy didáctico.
>
> Si un día tengo un tiempo muerto por ahí, veré si hay maneras de optimizarlo.

Sería genial, ya que en grandes mapas, el rendimiento es una variable importante. Si te sirve, un tip, dónde hice la parte de las direcciones, nunca quede demasiado conforme con la doble validación de Dir -> + Usar 8 Valores Anteriores para cálcular no volver para atrás. Siempre pensé habría una manera más prólija.

Saludos, Hernán . -

Por [Sheer] el 04 de Octubre de 2010

Muy interesante, de hecho hace tiempo estuve haciendo algo parecido, aunque mucho menos optimizado.

Por otra parte, me da la sensación que, al eliminar posiciones, das por hecho que el camino más rápido será hacia las posiciones que mantienes, pero pueden h obstáculos que hagan como camino más rápido o incluso único, el ir hacia las direcciones anuladas.

Vale, veo que has aclarado ésto en el primer comentario pero, aun y así, solo se habilitarian en caso de que las mantenidas no tengan solución, pero... ¿y si tien solución pero ir hacia atrás sería más rápido?

¡O quizá se me esté escapando algo!

Por Hernán el 04 de Octubre de 2010

[Sheer]-blog :

> Vale, veo que has aclarado ésto en el primer comentario pero, aun y así, solo se habilitarian en caso de que las mantenidas no tengan solución, pero... ¿y si tienen solución pero ir hacia atrás sería más rápido?
>
> ¡O quizá se me esté escapando algo!

Pues nunca ir para atrás es más rápido, menos si estamos estimando con Heurística los cálculos, que solo nos dirán la distancia del Punto B[Analizado] al Punto A. La lógica implica que si caminamos en dirección "correcta", siempre deberíamos obtener un valor más bajo de H, ergo, siempre saldrán esos valores elegidos, evitando el cálculo de 3 variables por interacción.

Please que alguien me corrija si estoy equivocándome en mi análisis.

Saludos, Hernán . -

Por daz_angie el 04 de Octubre de 2010

Hermosamente explicado.

Realmente lo expones como 4dummies.

Para hacerlo realmente inteligente, considera cada cuadro como un agente, para que se convierta en un sistema multiagente. Agregando un algoritmo de comunicación entre agentes y un algoritmo de asynchronous backtracking. Si consideras todo como restricciones dentro de las heurísticas, puedes lograr que resuelva un problema extremadamente complicado, aunque la capacidad de procesamiento necesaria es relativa a la complejidad y tamaño.

Esto es ya para un sitema inteligente más grande y con más complicaciones. Es parte de la Teoría de Juegos.

Por Hernán el 05 de Octubre de 2010

daz_angie-blog :

> Hermosamente explicado.
>
> Realmente lo expones como 4dummies.

JAJA gracias, intentaba justamente crear una atmosfera sencilla, estoy harto que todo lo que sea sinónimo de "interesante" sea igual a "no lo podrás leer"

daz_angie-blog :

> Es parte de la Teoría de Juegos.

Interesante idea, intentaré abordar la esquemática en siguientes tips. Aún me falta muuucho para llegar a eso je.

Por Biktor el 05 de Octubre de 2010

Muy bueno y encima bien explicado.
Gracias

Por daz_angie el 05 de Octubre de 2010

Hernán :

daz_angie-blog :

Es parte de la Teoría de Juegos.

Interesante idea, intentaré abordar la esquemática en siguientes tips. Aún me falta muuucho para llegar a eso je.

Lo que necesites de ese tema o AI dime. Me estoy especializando en eso

Por eveevans el 06 de Octubre de 2010

Creo q retomaste varios articulos q habias hecho >
bien Hernan, son genial ese tipo de articulos

Por Hernán el 06 de Octubre de 2010

eveevans-blog :

Creo q retomaste varios articulos q habias hecho >
bien Hernan, son genial ese tipo de articulos

JAJA no, solo tenía un tiempo y me puse a escribirlos jajaja..

Aún tengo varios en mente, pero mi semana no fue tan bien en tiempos como esperaba jaja

Por Mauro Cifuentes el 06 de Octubre de 2010

Saludos a todos, muy bueno el articulo, yo hice tambien una implementacion de A* en javascript, igual lo voy a analizar porque seguramente esta mucho mejor c
el mio (por ejemplo, el mio no se mueve en diagonal, porque no necesito que el objeto que quiero mover realice movimientos diagonales)
pueden verlo en accion en http://www.whitecat.com.ar/demos/astar

Por Kerly el 02 de Julio de 2011

No tendras el codigo completo

Por leomont el 10 de Abril de 2012

Podrias Explicarme de donde se calcula el valor de H?

Por leomont el 10 de Abril de 2012

SI G vale 10 para los ortogonales y 14 para los diagonales, por que es el mismo valor para todos?

Por fff el 20 de Julio de 2012

ffds

Por xd el 20 de Julio de 2012

alguien tiene el codigo completo y que funcione correctamente?

Por Lisandro el 04 de Noviembre de 2012

El ejemplo tiene 2 errores con lo que mencionás al principio.
G debería valer 10 para los ortogonales y 14 para los diagonales, y ahí en la gráfica figuran todos con 14, y el valor de H al principio mencionaste que había que multiplicar la distancia *10.
Para el primer movimiento quedaría en lugar de 19, 54(14+5*10) que sería el menor.

Por el 28 de Septiembre de 2016

Fashion is easiest to talk about when it's fun. Mulberry replica handbags that will be used in the evening or on weekends can be small, highly detailed, brightly colored--anything you want them to be! Most of us, though, spend a lot of times hauling around a bag that has to be large, functional and neutral, because that's v our professional lives require. Bags that match that description can be had at almost any price point, but in my mind, the sweet spot for most people interested in designer accessories is probably somewhere in the three figures. That's enough to get you a beautiful replica handbags that feels well-made and luxurious, but it' also reasonable enough that it might not blow your budget for something a little more fun to use in your off-hours. Below, check out ten bags in that range that we totally love. Celebs seem to be playing a game of musical chairs with designer brands this week. All of our top A-list handbag lovers have abruptly switched out th summer favorites for something new. Reese is carrying Chloé replica handbags, Miranda is carrying Bulgari and Stranger Things star Millie Bobby Brown has ma her very first appearance on our virtual pages. We haven't done a version of "Many Bags" for celeb children and pre-teens yet, but it could be forthcoming. First u we have Bella Hadid, leaving an establishment called Sexy Fish (your guess is as good as mine) with a dainty silver Dior Diorama. This Diorama is a very recent addition to Bella's miu miu outlet handbag collection, which you can see it in full in The Many Bags of Bella Hadid.

Por el 31 de Enero de 2018

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